

Quiz #20

Create a function (vec.m) that will calculate and return the position vector and the magnitude of the position vector given two coordinate points (A and B). The function should also calculate and return the unit vector if required. The script that calls the function should use the menu function to ask the user if they want the unit vector to be calculated. The function header is shown below – do not change the function header.

```
function [ r,varargout ] = vec( A,B )
```

Use the following coordinates to test your code:

A=[2, 3, 4]

B=[4, 6, 9]

Recall: $\vec{r} = A - B$

$$\hat{u} = \frac{\vec{r}}{r}$$

